

1.5T CTL Coil

4 Channel CTL Coil

8 Channel CTL Coil

Operator Manual

2422287-1EN



Operator Manual for the 1.5T CTL Coil
2422287-1EN
Revision 2 (2018-03)



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Revision History

Revision	Date	Author	Primary Reason for Change
01	17-Apr-2013	D Seaman	Initial Release
02	28-Mar-2018	D Seaman	Page All: updated revision, date, copyright date, changed GE Healthcare to General Electric Company in footer, changed caution statement symbol to yellow background Title page: spelled out the word "revision", updated revision, date, copyright date, removed 'General Electric doing business as' statement, added CE mark, removed "GE Healthcare" Page i: added revision table Page ii: added "The year of CE Marking..." sentence, moved Intended Use section to page 2-1 of Chapter 2 Page iii: added eIFU section Page 1-1: removed "Healthcare" from "For questions or comments..." sentence in the Please review this manual..." paragraph Page 1-6: added second receive only symbol and added caution symbol Page 1-8: updated fragile and keep dry symbols Page 2-5: removed "Healthcare" from "Any changes or modifications..." sentence of the Modifications section, from fourth bullet point on page, and from "Users should immediately..." sentence Page 2-6: removed "Healthcare" from "If you need additional training..." sentence of first paragraph of MR Safety Information section Page 3-3 to 3-8: updated figures 3-1 to 3-12 for translation Page 3-11: updated photograph aligning with steps 3 and 4 Page 3-13: updated figure 3-13 Page 4-1: removed "Healthcare" and phone number from "Please contact GE..." sentence in the Quality Assurance section Page 4-3: clarified 70% isopropyl alcohol section of first bullet point, removed "Healthcare" from second bullet point and removed "Healthcare" and phone number from "If you do not have..." sentence in the Accessories and Replaceable Components section

Medical Device Directive

This product conforms with the requirements of council directive 93/42/EEC concerning medical devices when they bear the following CE Mark of Conformity:



The year of CE Marking is 1998.

EU Authorized Representative:

GE Medical Systems S.C.S.

Quality Assurance Manager

283 rue de la Minière

78530 BUC France

Telephone: +33 1 30 70 40 40

Manufacturer:

GE Healthcare Coils

1515 Danner Drive

Aurora, Ohio 44202-9273



CAUTION:

This symbol indicates that the waste of electrical and electronic equipment must not be disposed as unsorted municipal waste and must be collected separately. Contact an authorized representative of the manufacturer for information concerning the decommissioning of your equipment.



CAUTION:

United States federal law restricts the sale, distribution and use of this device to, or on the order of, a physician.

Electronic Instructions for Use



Your MR device is delivered with the electronic Instructions For Use (eIFU) logo. The electronic Instructions For Use (eIFU) symbol means that the operator manual can be viewed from a PC.

Refer e-User Leaflet 5452943-199 for further instructions.

The latest version of the Instructions for Use is available on the Internet at: <http://apps.gehealthcare.com/servlet/ClientServlet?REQ=Enter+Documentation+Library>

On the home page enter 2422287-399 in the search window and launch the search.

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Chapter 1

Introduction and Coil Specifications

Introduction

This manual describes the safety precautions, features, use and care of the following receive-only 1.5T CTL Coils compatible with the systems listed:

- Premier 7000 QD CTL PA Coil (Receive Only) (4 channel) - GE 1.5T MRI Systems with Legacy Bendix Connector
- 8 Channel CTL Coil - GE 1.5T MRI Systems with Legacy Bendix Connector
- Signa HD 1.5T CTL Spine Coil (4 Channel) - GE 1.5T MRI Systems with HD Connector
- 1.5T HD CTL Array (8 channel)- GE 1.5T MRI Systems with HD Connector

Throughout this manual the 4 Channel Premier 7000 QD CTL PA Coil (Receive Only), the 8 Channel CTL Coil, the 4 Channel Signa HD 1.5T CTL Spine Coil, and the 1.5T HD CTL Array will be referred to by their individual names, or as the 1.5T 4 Channel CTL coil and the 1.5T 8 Channel coil or collectively as the 1.5T CTL coil.

Please review this manual thoroughly before using the device for imaging. Refer to your MR system manual in conjunction with this manual. For questions or comments on this manual, or assistance with the use of this product, please contact a GE representative.

This equipment is for use by qualified personnel only.

NOTE: This manual should be kept with the equipment and available at all times. It is important to periodically review the procedures and safety precautions. **All users must read and understand the contents of this manual before attempting to use this product.**

Premier 7000 QD CTL PA Coil (Receive Only) (4 Channel) Shipping Contents

NOTE: Shipping contents may be packaged in more than one carton.

Table 1-1 Premier 7000 QD CTL PA Coil (Receive Only) (4 Channel) Shipping Contents:

Part Name	Part Number	Qty.
Premier 7000 QD CTL PA Coil (Receive Only)	U1-100082	1
Patient Pad	2423521	1
Wedge/Ear Pads	2423515	2
Adapter Block	U1-150038	1
Operator Manual Packet	2422287-500	1
Service Manual	2225545-11	1
C-Spine Bridge, GE 1.5T	U1-110004	1

8 Channel CTL Coil Shipping Contents

NOTE: Shipping contents may be packaged in more than one carton.

Table 1-2 8 Channel CTL Coil Shipping Contents:

Part Name	Part Number	Qty.
8 Channel CTL Coil	U1-100341	1
Wedge/Ear Pads	2423515	2
Knee Pad	2423517	1
Comfort Pad	2423513	1
Adapter Block	U1-150038	1
Operator Manual Packet	2422287-500	1
Service Manual	2377426-5	1

Signa HD 1.5T CTL Spine Coil (4 Channel) Shipping Contents

NOTE: Shipping contents may be packaged in more than one carton.

Table 1-3 Signa HD 1.5T CTL Spine Coil (4 Channel) Shipping Contents:

Part Name	Part Number	Qty.
Signa HD 1.5T CTL Spine Coil	2415369	1
Patient Pad	2423521	1
Wedge/Ear Pads	2423515	2
Adapter Block	U1-150038	1
Operator Manual Packet	2422287-500	1

1.5T HD CTL Array (8 Channel) Shipping Contents

NOTE: Shipping contents may be packaged in more than one carton.

Table 1-4 1.5T HD CTL Array (8 Channel) Shipping Contents:

Part Name	Part Number	Qty.
1.5T HD CTL Array	2415375	1
Wedge/Ear Pads	2423515	2
Knee Pad	2423517	1
Comfort Pad	2423513	1
Adapter Block	U1-150038	1
Operator Manual Packet	2422287-500	1

Use of Symbols

Coil users should understand the following symbols used for caution and warning explanations:



CAUTION:

Indicates a hazard that may cause a minor injury or property damage if instructions are not followed.

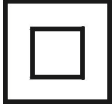


WARNING:

Indicates a hazard that may cause severe personal injury, death, or substantial property damage if instructions are not followed.

Coil Classification

Table 1-5 Classification of 1.5T CTL Coil:

Symbol	Classification
	Class II equipment
	Type BF applied part

- Suitable for continuous operation.
- The IP Rating for the coil is IPX0. Ordinary equipment. Not rated against water ingress protection.
- Non-sterile device.
- Equipment not suitable for use in the presence of a flammable anaesthetic mixture with air or with oxygen or nitrous oxide.
- The voltage rating of the coil is 10VDC maximum.

All surface areas of the coil, except the bottom surface that sits on the patient table, are considered applied parts and may come into direct contact with the patient. Pads and/or thermal-resistant materials should be used to prevent the coil surfaces from touching the patient.

Pads and/or thermal-resistant material must be used to ensure the patient cannot touch the coil cable and connectors.

Symbol and Label Definition

Table 1-6 Classification of 1.5T CTL Coil:

Symbol	Classification
	Manufacturer
	Date of Manufacture
	Consult Accompanying Documents
	Consult Accompanying Documents
	Serial Number
	Part Number
	Cables bearing this ID label are compatible with software versions 10.x and newer. On software versions 11.x and newer, the system will generate an error message when an incompatible coil is connected to the system.
	Landmarking labels (arrow) run down both sides of coil
	1.5T 8 Channel CTL Coil element section markings on pad. Markings are on both sides of the pad.

Table 1-6 Classification of 1.5T CTL Coil: (Continued)

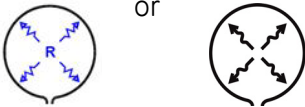
Symbol	Classification
	Do Not Loop the Cable Symbol
	Surface May Be Hot
	Do not loop the cable warning label
	This symbol indicates that the waste of electrical and electronic equipment must not be disposed as unsorted municipal waste and must be collected separately. Contact an authorized representative of the manufacturer for information concerning the decommissioning of your equipment.
 or	Receive Only Coil
	Indicates a hazard that may cause a minor injury or property damage if instructions are not followed.
	1.5T 8 Channel CTL coil label
	Plug in the connector
	1.5T CTL coil Anterior Bridge Label

Table 1-6 Classification of 1.5T CTL Coil: (Continued)

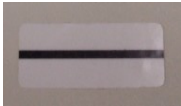
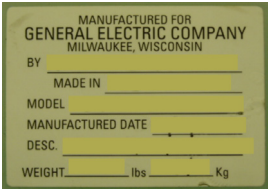
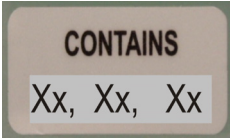
Symbol	Classification
	1.5T CTL coil Anterior Bridge part number
	1.5T CTL coil Adapter Field Replaceable Unit (FRU) number
	1.5T CTL coil Phantom centerline label. Label is on the coil.
	1.5T CTL Coil label
	Phantom Identifier
	Phantom Contents
	Phantom positioning label. Label is on the phantom.
	Packaging CE label

Table 1-6 Classification of 1.5T CTL Coil: (Continued)

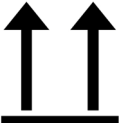
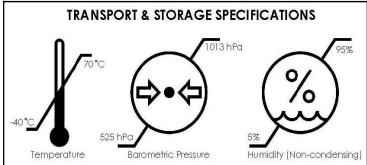
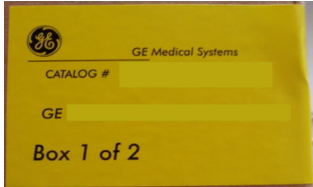
Symbol	Classification
	Fragile - Do not drop or handle package roughly
	Keep the packaging out of the rain. Keep packaging dry.
	This end up. The arrows point to the side of the package that is to remain facing up.
	Legacy Do not allow package to freeze
	Legacy Do not drop or handle package roughly
	Legacy Label. Keep packaging dry. Do not drop or handle package roughly. The arrow points to the side of the package that is to remain facing up.
	Transportation and Storage Specifications Label. The symbol on the left communicates the temperature specifications. The symbol in the middle communicates the atmospheric pressure specifications. The symbol on the right communicates the humidity specifications.

Table 1-6 Classification of 1.5T CTL Coil: (Continued)

Symbol	Classification
 Transportation & Storage Specifications 1. Barometric Pressure: 645 mm Hg to 795 mm Hg 2. Humidity: 20% to 95% non-condensing 3. Temperature: -15 °C to 55 °C USA Instruments, Inc.	Legacy Transportation and storage specifications
 GE Medical Systems CATALOG # GE Box 1 of 2	Shipping content label. Identifies the catalog number, GE part number and the number of boxes in the shipment.
 PO # PN Quantity Shipping Slip Number Serial Number Date Shipped USA Instruments, Inc. 1515 Dunbar Drive Aurora, Ohio 44202, U.S.A.	Identifies contents, purchase order number, part number, quantity, shipping slip number, serial number and date shipped.
 Manufactured By USA Instruments, Inc. 1515 Dunbar Drive Aurora, Ohio 44202, U.S.A. PO NUMBER SHIP DATE MXXXXX 1 of 2	Shipping label. Identifies manufacturer, purchase order number, serial number, ship date, part numbers and number of boxes in the shipment.

Coil Description



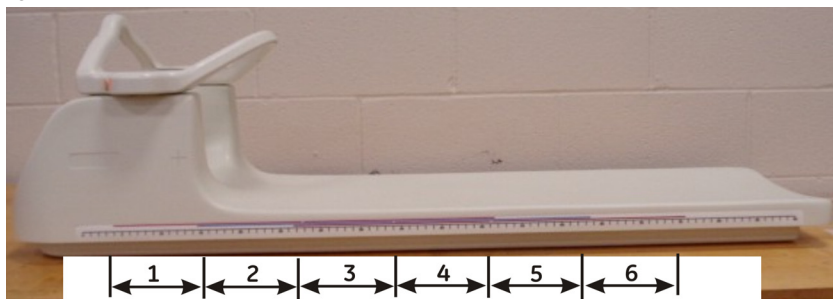
The 1.5T 4 Channel and 8 Channel CTL Coils are receive-only coils designed to give optimum signal-to-noise ratio and uniform coverage of the spine anatomy, including the cervical, thoracic and lumbar regions. Each coil has a removable cervical bridge for enhanced imaging of the anterior cervical region.

The 1.5T CTL coil allows the user to obtain high-resolution images of extended regions (such as the cervical-thoracic spine and thoracic-lumbar spine)

or specific regions of interest (such as the cervical spine).

Each 1.5T CTL coil has six sections in the head-to-foot direction. There are a total of 12 elements; each coil section includes two elements and uses two channels through the interface. Twelve coil modes use a combination of these elements.

Figure 1-1 1.5T CTL Coil Element Sections:



The 1.5T 4 Channel CTL Coils and the 1.5T 8 Channel CTL Coils are designed to be used by a Registered MRI Technologist. The Registered MRI Technologist will operate the scanner from the control room. If the patient is claustrophobic another MRI Technologist or clinical staff member may stay in the magnet room with the patient.

Coil Specifications

Table 1-7 CTL Coil Specifications:

Field Strength	Model	Part #	Frequency
1.5T	Premier 7000 QD CTL PA Coil (4 Channel)	U1-100082	63.86 MHz
1.5T	Signa HD 1.5T CTL Spine Coil (4 Channel HD)	2415369	63.86 MHz
1.5T	8 Channel CTL Coil	U1-100341	63.86 MHz
1.5T	1.5T HD CTL Array (8 Channel)	2415375	63.86 MHz

Coil Dimensions and Weight

NOTE: Store the coil in an area that exceeds the dimensions of the coil.

Dimensions

110cm Long x 40.9cm Wide x 34.5cm High (43.3in Long x 16.1in Wide x 13.6in High)

Weight

9.1kg (20 lb)

Coil Installation and Service

Coil installation and configuration must be completed by a GE Service Representative. Please contact a GE Service Representative for detailed coil installation and configuration information. The GE Field Engineer shall consult the appropriate service manual or service method CD for installation instructions.

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Chapter 2

Safety

Patient safety and comfort are the primary concerns during the scanning procedure. Always follow proper safety, operating and maintenance procedures to ensure that the patient is not exposed to electrical or mechanical hazards that may cause injuries.

Route cables through the center of the magnet bore. Routing cable near the sides of the bore increases the likelihood of cable heating from induced currents. Place cables under the cushion whenever possible.

Keep the length of the cable in the bore to a minimum. When possible, avoid bending the cable 180 degrees. Route cables out of the bore in the most direct way possible, without looping or coiling.

Ensure that the patient is comfortably positioned.

Monitor the patient periodically. If the patient reports heating, burning or tingling sensations, **stop the scan immediately**.

All personnel using this coil must be instructed in its proper use. Personnel must observe all warnings and cautions appearing in this manual.

Intended use for the 1.5T CTL Coil

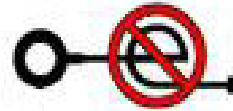
The 1.5T Phased Array CTL Spine Coil (1.5T CTL coil) is a receive-only phased array RF coil, used for obtaining diagnostic images of the spine region (cervical, thoracic, and lumbar anatomy) in Magnetic Resonance Imaging systems.

Patient Safety



WARNING:

Patient burn risk. Do not cross or loop coil cables. Do not loop RF receive coil cables and ECG leads.



WARNING:

Do not allow the coil cable to come into contact with the skin of the patient. Use a thermal resistant material or pad to keep the cable from touching the patient. Failure to do so may result in patient burns.



WARNING:

RF can cause localized heating at contact points between adjacent body parts when a loop is formed. Such localized heating can result in a tingling sensation, discomfort, or burns. This could occur when a patient's hands are touching or when a female patient's breasts are compressed to her chest. Use pads between body parts to avoid creating a loop with adjacent body parts.



WARNING:

Do not use accessories (e.g. pads or straps) that have not been specifically tested and approved for use in the MR environment. Use only GE recommended accessories. Use of non-approved accessories may result in patient burns or injuries or image degradation. Even auxiliary devices labeled as compatible with MR equipment are capable of causing injury if the manufacturer's instructions are not followed.



WARNING:

RF can cause localized heating at contact points between the patient and the bore resulting in discomfort or burns. Place appropriate non-conductive padding between the patient and the bore wherever a portion of the body may come into contact with the magnet opening.



WARNING:

Closed loops formed by clasped hands or crossed arms, legs or feet may cause burns to the patient. Do not allow the patient to cross or loop their hands, arms, legs or feet. Use pads as necessary to separate limbs.



WARNING:

Always place appropriate non-conductive padding between the surface coil and the skin of the patient to prevent burn injuries.



WARNING:

Prior to patient placement in the coil, assure that any breached or compromised patient skin surfaces that come in contact with the coil have been adequately bandaged or covered.



CAUTION:

To avoid injuring the patient or damaging the coil, watch carefully as the table moves into the bore. Stop advancing the table if the patient or any part of the coil comes into contact with the bore.

Equipment Safety



WARNING:

Auxiliary equipment (e.g. gating equipment, vital signs monitoring systems and RF coils) that has not been specifically tested and approved may result in burns or other injuries to the patient. Such equipment may also interfere with proper operation of this coil. Only use equipment approved for the MR environment.



WARNING:

This coil is not suitable for use in the presence of flammable anesthetic mixture with air or with oxygen or nitrous oxide.



WARNING:

Patient burns may result from RF coupling of this device with other devices remaining in the magnet. Remove any unused accessory device, coil, or cable from the magnet before using this coil.



WARNING:

Route cables through the center of the magnet bore. Place cables under the pads when possible. Routing cables near the sides of the bore increases the likelihood of cable heating (from induced currents).



CAUTION:

RF coupling risk. Looped cables may cause RF coupling and degrade the scan performance of the coil. Do not cross or loop coil cables.



Electrical and Mechanical Safety



WARNING:

Electric shock hazard. This coil consists of electrical and mechanical components. Tampering with the coil by untrained personnel can be hazardous to the patients and equipment. Only properly trained and qualified personnel should service the coil.



WARNING:

Use only coils, cables and accessories that are in good condition. Before using the coil, visually check each coil component, cable and accessory to ensure that there is no external damage. If any coil component, cable or accessory is suspected of not being in good condition, discontinue use and contact a GE Service Engineer.



WARNING:

Electric shock hazard. No user serviceable parts. Refer service to qualified service personnel.

Definitions for Labels Located on the Coil



WARNING:

The label shown in the photograph to the right is of a Do Not Loop Cable warning. It is located on the interface box of coils with HD connectors. This label instructs the user to not loop or cross coil cables.



WARNING:

The label shown to the right can be found on the coil cable. It contains three symbols: surface may be hot, do not loop, and consult accompanying documents.



The serial number label for the coil also contains several symbols. This label specifies date of manufacture, place of manufacture, part number and revision. This label also has several symbols that are defined in this manual that will help the user better understand the specifics of the coil.

Modifications

Any changes or modifications to this coil must be approved and performed by GE prior to installation.

Emergency Procedures and Incident Reporting

If a patient requires emergency assistance:

- Stop the scan if one is in progress.
- Remove the patient from the scan room if medical treatment is needed.
- Please refer to MR Safety chapter in your MR system operator manual or refer to the MR Safety Guide, part #2381696, if you received one with your system, for patient emergency procedures.

In the unlikely event that a coil creates smoke, sparks or makes an unusually loud noise:

- Contact GE to report the incident.
- Do not use the coil until a GE Service Representative inspects the coil and indicates it is safe to use.

Users should immediately report to GE any incidents or injuries that result from coil operation involving a patient, operator, or maintenance employee.

If an accident occurs as a result of coil operation, do not operate the equipment until an authorized investigation has been conducted.



CAUTION:

Users should place a service call any time the coil is dropped or mishandled. A GE Service Representative should inspect the coil after it has been dropped or mishandled to ensure the coil is safe to use.

MR Safety Information

Please refer to MR Safety chapter in your MR system operator manual or refer to the MR Safety Guide, part #2381696, if you received one with your system. The MR Safety documentation describes the safety information you and the physicians must understand thoroughly before you begin to use the MR system. If you need additional training, seek assistance from qualified GE personnel.

This equipment is intended for use by qualified personnel only.

This manual should be kept with the equipment and be readily available at all times. It is important for you to periodically review the procedures and safety precautions. **It is important for you to read and understand the contents of this manual before attempting to use this product.**

Chapter 3

Coil and Patient Setup

Lifting and Transporting the Coil

- Pick up the coil by firmly gripping the head and foot ends of the coil. There is a recessed area at the foot end of the coil that can be used to help lift the coil.
- Support the coil from underneath with two hands when carrying the coil.



Do not attempt to pick up the coil with the cervical bridge when the bridge is attached to the coil. Do not carry the coil by the cable. The coil may not work if damaged.

CAUTION:

Coil Mode Configurations

To optimize image quality using the 1.5T CTL coil, select a coil mode appropriate to the anatomy of interest and the desired FOV. Improper use of the coil is the major cause of image artifacts.

Table 3-1 Modes of Coil Operation:

1.5T Premier 7000 QD (4 Channel) Coil Modes	1.5T 4 Channel HD CTL Coil Modes	1.5T 8 Channel and 1.5T HD CTL Coil Modes	Clinical Use	Selected Sections	Figure Reference
USCTLTOP	USCTLTOP	8CTLTOP	Cervical and thoracic spine	1, 2, 3, 4	3-1
USCTLMID	USCTLMID	8CTLMID	Thoracic and lumbar spine	2, 3, 4, 5	3-2

Table 3-1 Modes of Coil Operation: (Continued)

1.5T Premier 7000 QD (4 Channel) Coil Modes	1.5T 4 Channel HD CTL Coil Modes	1.5T 8 Channel and 1.5T HD CTL Coil Modes	Clinical Use	Selected Sections	Figure Reference
USCTLBOT	USCTLBOT	8CTLBOT	Thoracic and lumbar spine	3, 4, 5, 6	3-3
USCS123	USCTL123	8CTL123	Cervical and thoracic spine	1, 2, 3	3-4
USCT234	USCTL234	8CTL234	Thoracic spine	2, 3, 4	3-5
USTL345	USCTL345	8CTL345	Thoracic and lumbar spine	3, 4, 5	3-6
USLS456	USCTL456	8CTL456	Thoracic and lumbar spine	4, 5, 6	3-7
USCS12	USCTL12	8CTL12	Cervical spine	1, 2	3-8
USTS23	USCTL23	8CTL23	Thoracic spine	2, 3	3-9
USTL34	USCTL34	8CTL34	Thoracic spine	3, 4	3-10
USLS45	USCTL45	8CTL45	Thoracic and lumbar spine	4, 5	3-11
USLS56	USCTL56	8CTL56	Lumbar spine	5, 6	3-12

Figure 3-1 Eight coil elements are active for this mode:

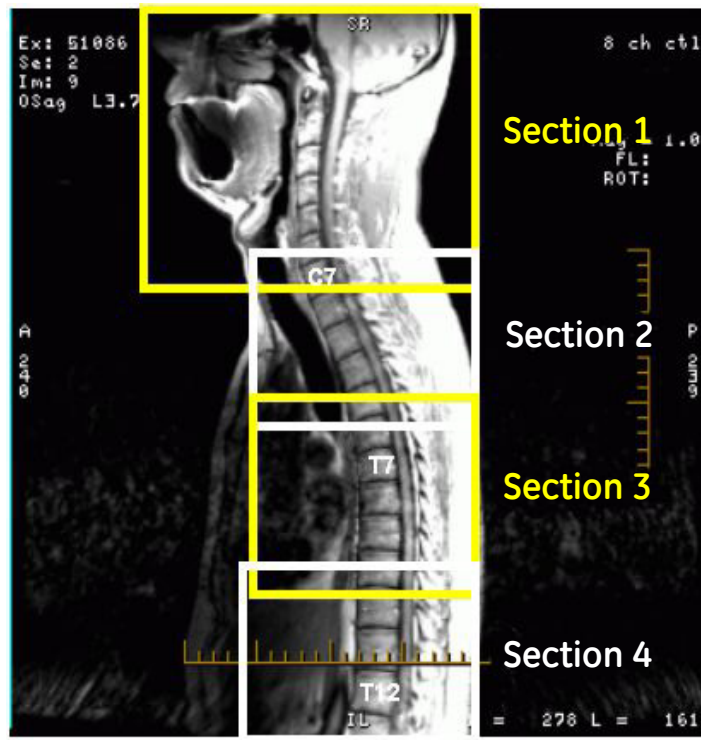


Figure 3-2 Eight coil elements are active for this mode:

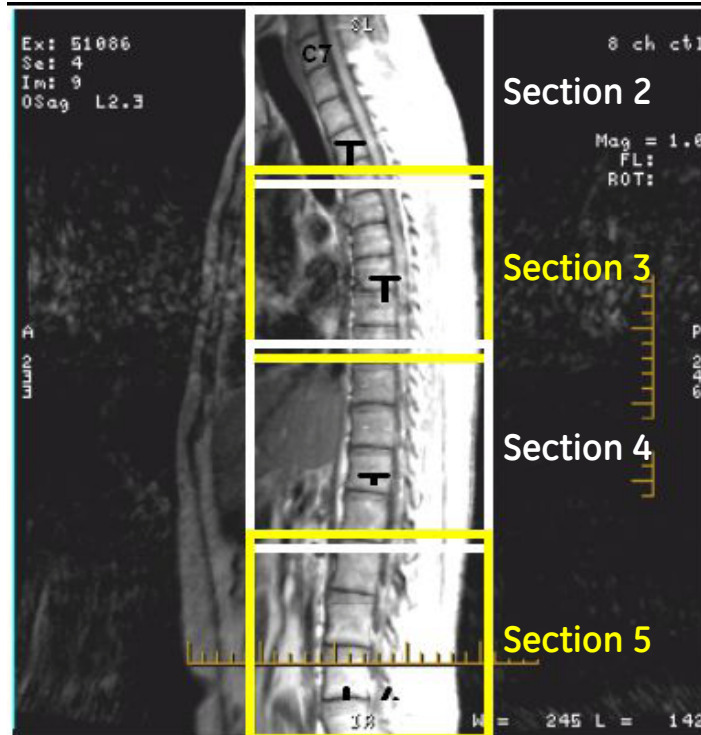


Figure 3-3 Eight coil elements are active for this mode:

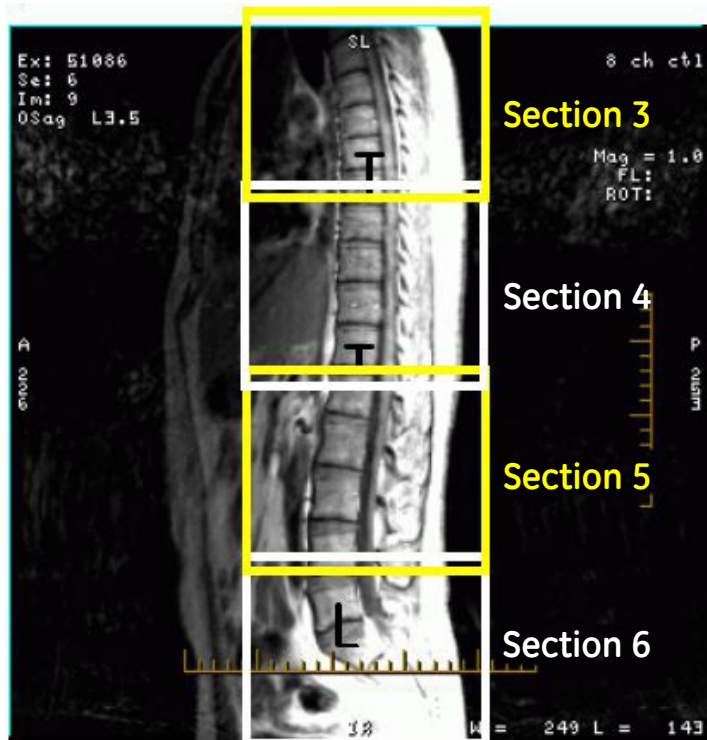


Figure 3-4 Six coil elements are active for this mode:

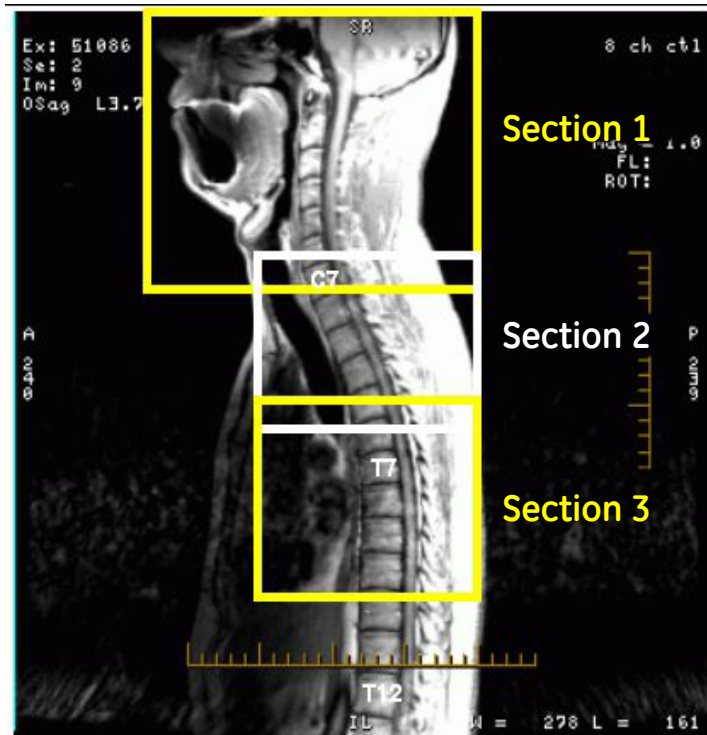


Figure 3-5 Six coil elements are active for this mode:

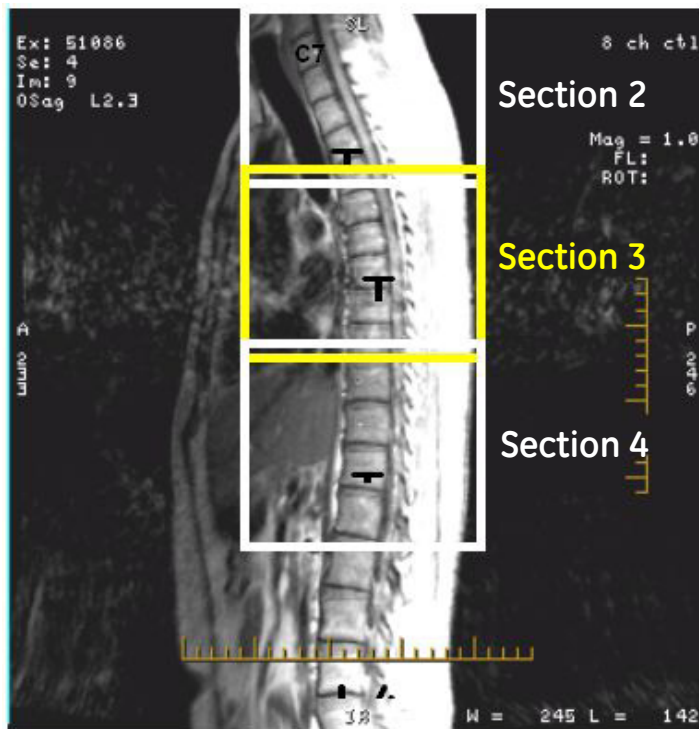


Figure 3-6 Six coil elements are active for this mode:

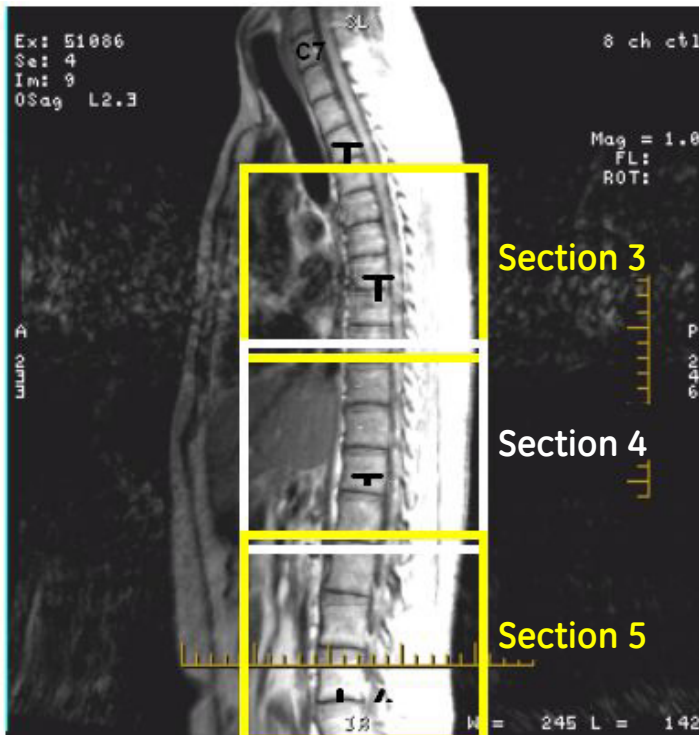


Figure 3-7 Six coil elements are active for this mode:



Figure 3-8 Four coil elements are active for this mode:

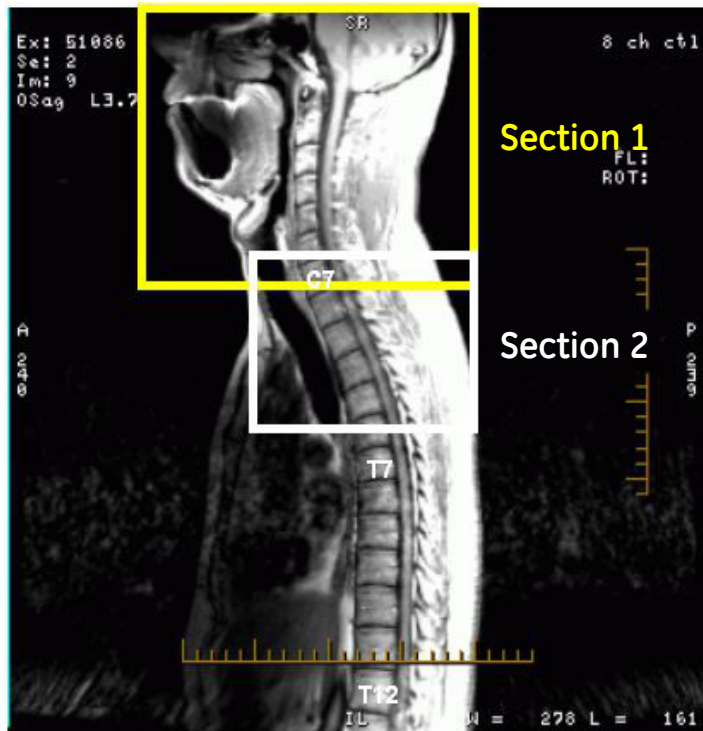


Figure 3-9 Four coil elements are active for this mode:

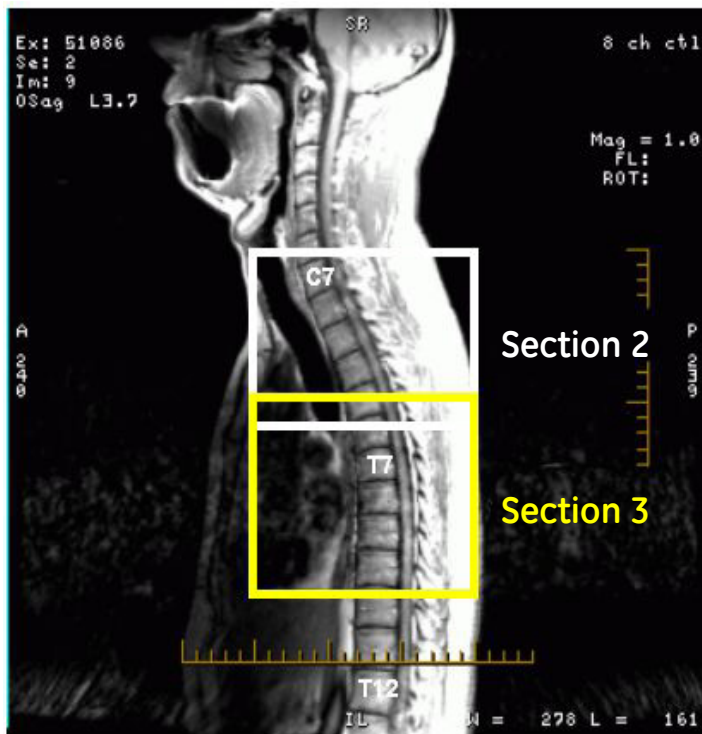


Figure 3-10 Four coil elements are active for this mode:

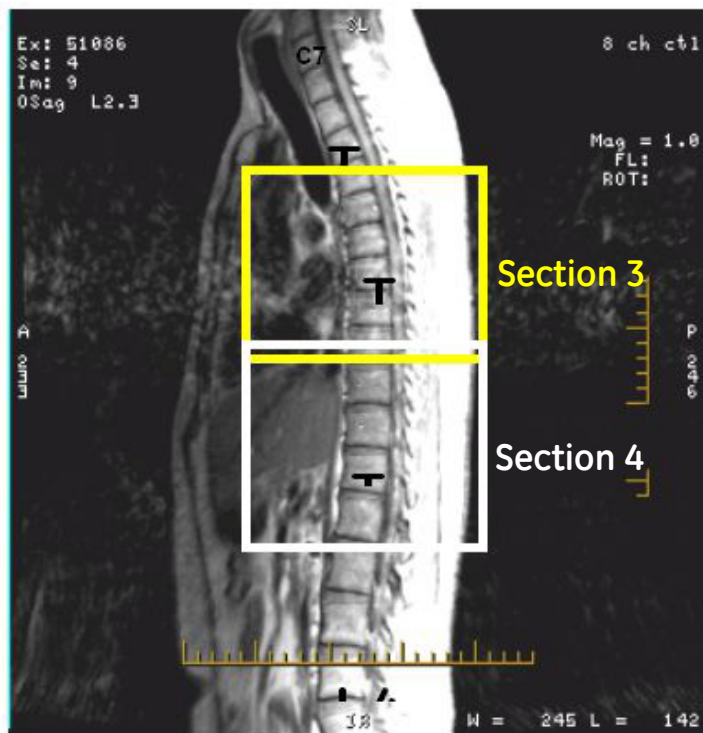


Figure 3-11 Four coil elements are active for this mode:

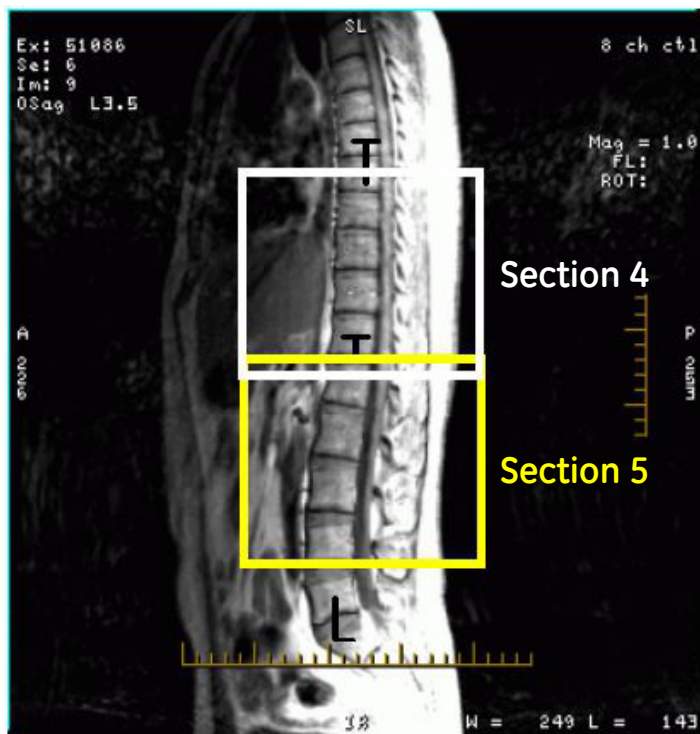


Figure 3-12 Four coil elements are active for this mode:



Positioning the Coil and Patient



WARNING:

RF coupling danger. Patient burns may result from RF coupling of this device with other devices remaining in the magnet. Remove any other coil or unused accessory device from the magnet before using this coil.



WARNING:

Patient burn risk. Do not cross or loop coil cables. Do not allow any cable to touch the patient. Use a thermal-resistant material or pad to prevent cables from touching the patient.



CAUTION:

Be sure the coil is unplugged before lowering the table. Failure to do so may damage the coil, cable, or connector. The coil may not work if damaged.

Placing the Coil on the Table

Table 3-2 Position the Coil on the Scan table:

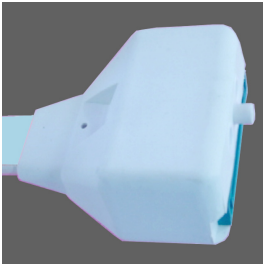
Visual	Description
	Position the coil (with the pad) on the table near the bore of the magnet.

Head-First Imaging

Table 3-3 Positioning Patient on the Scan Table for Head-First Imaging:

Visual	Description
	1. Remove the cervical bridge of the coil.
 CAUTION:	Equipment damage risk. Improper removal of the cervical bridge will cause coil damage. Press both release buttons simultaneously to remove the bridge.
 	2. For best image quality, the cervical bridge should be used. However, if patient size does not allow for use of the cervical bridge, attach the adapter block to the coil.
 WARNING:	Patient burn risk. Leaving the connector exposed may cause burns or shock to the patient if touched during the scan. Always use the cervical bridge or adapter block when scanning with the CTL coil.

Table 3-3 Positioning Patient on the Scan Table for Head-First Imaging: (Continued)

Visual	Description
	3. Position the patient head first, supine, with the top of the shoulders resting firmly against the coil sides. Use earpads to secure the patient's head in the coil. Position the patient's arms to his or her sides. NOTE: A cotton sheet should be used as necessary to keep the patient's skin from touching the coil. 4. Align the cervical bridge with the alignment pegs and press the bridge onto the posterior section of the coil.
 4 channel Legacy Bendix  8 channel Legacy Bendix  HD Connector	5. Plug the coil cable into the appropriately labeled port of the system receptacle. NOTE: The coil can be connected to the 4 channel or 8 channel Legacy Bendix ports or the A, B or C HD ports depending on the coil type and the system configuration. NOTE: The coil connector and cable should always be routed directly to the magnet and should not form loops. 6. Localize the anatomy and advance to scan.

Feet-First Imaging

NOTE: Feet-First imaging is **Not** recommended for the 4 Channel CTL coil. Only use the feet-first imaging position with the 8 Channel CTL coils and the knee transition pad included with the coils.



Patient burn risk. Leaving the connector exposed may cause burns or shock to the patient if touched during the scan. Always use the adapter block when using the knee transition pad for feet-first scanning.

WARNING:

Table 3-4 Positioning the Patient on the Scan Table for Feet-First Imaging (8 Channel coils only):

Visual	Description
 WARNING:	Patient injury risk. When using the knee transition pad, ensure that the patient's knees do not touch the top of the bore. Burns or abrasions may occur if the knees touch the wall of the bore during scanning. Place appropriate non-conductive padding between the knees and the bore to prevent injury.
	1. Remove the cervical bridge and attach the adapter block. NOTE: Follow the instructions shown in steps 1 and 2 of Table 3-3 for proper removal of the cervical bridge and attachment of the adapter block. 2. Position the knee transition pad over the cervical end of the coil.
	3. Position the patient feet first, placing the legs snugly against the knee transition pad. Position the patient's arms to his or her sides. NOTE: A cotton sheet should be used as necessary to keep the patient's skin from touching the coil.

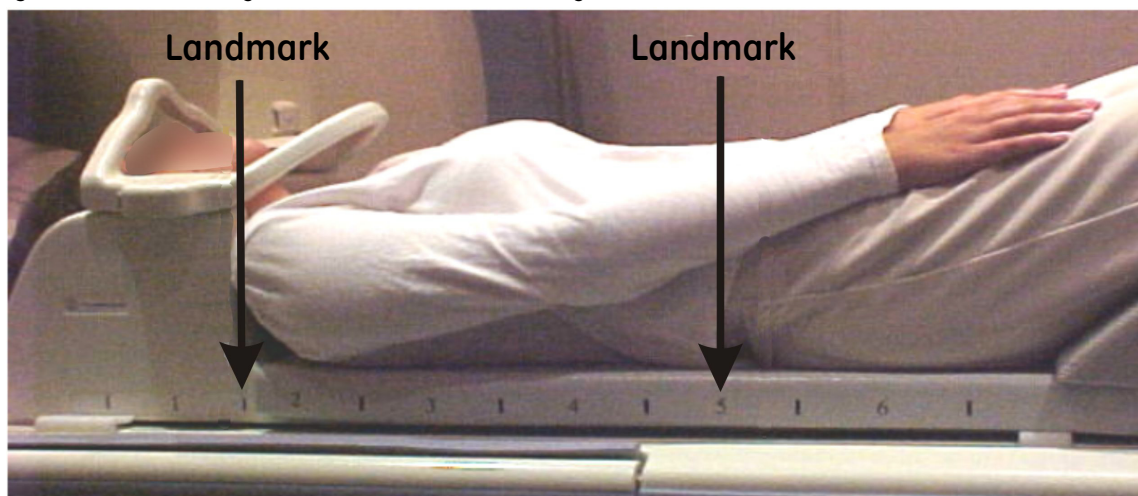
Table 3-4 Positioning the Patient on the Scan Table for Feet-First Imaging (8 Channel coils only):

Visual	Description
 8 channel Legacy Bendix  HD Connector	4. Plug the coil cable into the appropriately labeled port of the system receptacle. NOTE: The coil can be connected to the 8 Channel Legacy Bendix ports or the A, B or C HD ports depending on the coil type and the system configuration. NOTE: The coil connector and cable should always be routed directly to the magnet and should not form loops. 5. Localize the anatomy and advance to scan.

Localize Anatomy and Choose Mode Configuration

Localize Anatomy

Position the anatomical area of interest to the center of the coil section used. Base this positioning on the desired FOV. Landmark the location (Figure 3-13).

Figure 3-13 Landmarking for Cervical or Lumbar Scanning:

NOTE: Due to patient height, landmarking locations may not be exactly as shown.

Choose Mode Configuration

Refer to the beginning of this chapter, **Chapter 3 - Coil Mode Configurations**, for detailed descriptions of each coil mode configuration. Choose the configuration most appropriate for the corresponding anatomy of interest.

Each CTL Coil is provided with external marks that represent each coil section.

- Select the appropriate number of sections according to the area that needs to be covered. Considering that each section is 12cm long in the head-to-foot direction, select a coil configuration and FOV based on the number of sections chosen.
- Landmark on the coil external landmark, not on the anatomy.

NOTE: To reduce artifacts in sagittal or coronal imaging, use the minimum number of coil sections to image the given FOV. For any 4-section coil configuration, the maximum FOV is 48cm. For any 3-section configuration, the maximum FOV is 36cm. For any 2-section configuration, the maximum FOV is 24cm.

NOTE: Auto-N-Coil is only compatible with 4 Channel CTL coils, and will disable elements in the middle of a scan series to reduce undesired image artifacts.

Recommended Scan Protocols

Use the GE recommended protocols for scanning with this coil for optimum results and best performance.

Specific choice of protocols is entirely the responsibility of the radiologist and/or technologist.

Chapter 4

Coil Maintenance

Coil Maintenance - Introduction

To maintain the highest possible image quality and to protect patients and personnel from potential injury or illness, it is important to keep the coil clean and well maintained. This chapter lists basic methods for keeping the coil safe and sanitary. Also included is a list of accessories that will enhance the utility of your coil. Refer to the contact information for questions or comments.

Quality Assurance

Coil Quality Assurance tests must be run by a GE Service Representative or a third party service provider. To have a Quality Assurance test run on a coil, please call your GE Service Representative or your third party service provider.

If you do not have a GE Service Representative or third party service provider, use the Class A Service CD that was shipped with your MR System. This CD contains information on phantom setup and the usage of the MCQA tool. Please contact GE with any questions or concerns.

Environmental Requirements

Table 4-1 Coil Environmental Requirements:

Environmental Factor	Transport Conditions	Operating and Storage Conditions
Barometric pressure	645 mm Hg to 795 mm Hg (860 hPa to 1060 hPa)	As established by the GE recommended MR suite conditions.
Humidity	20% to 95% non-condensing	
Temperature	-15°C to 55° C (5° F to 131°F)	

Coil Storage

When the coil is not in transport and is not being used, store the coil in an air conditioned area that has environmental conditions that fall within the established GE recommended MR suite conditions.

The coil must be stored in a storage space that exceeds the dimensions of the coil.

Coil Care

The coil must be inspected before each use. Visually check each coil component, cable and accessory to ensure that there is no external damage. If any coil component, cable or accessory is suspected of not being in good condition, discontinue use and contact a GE Service Engineer.

All FRU's (Field Replaceable Units) must be replaced by a GE Service Engineer or a third party service provider. Component part lists with descriptions, instructions for the safe replacement of components, circuit diagrams, and calibration and preventive maintenance instructions are available to service personnel upon request.



WARNING:

Electric shock may occur if the coil is attached to the system during cleaning or when it is still wet. Detach coil connector from the scanner before attempting to clean the coil. Do not touch connectors with bare fingers. Never press sharp objects against connector surface. Do not reattach connector after cleaning until the coil has dried completely.

**CAUTION:**

The coil contains sensitive electronic components that may be damaged by liquids. Do not spray or pour cleaning solution directly onto the coil. Do not submerge the coil in any solution. Under no circumstances should the coil be placed into any type of sterilizer.

The 1.5T CTL Coil and patient pads must be thoroughly cleaned after each use.

- Wipe with a cloth that has been dampened in a solution of 10% bleach and 90% tap water, or with a 70% isopropyl alcohol solution made from 100% isopropyl alcohol and 30% tap water or utilize a commercially prepared solution of 70% concentration isopropyl alcohol.
- Should your coil need to be returned to GE for service, wipe it thoroughly with a 10% bleach solution (as described above) to minimize risk of exposure to potentially infectious agents.
- Dispose of any materials used to clean the coil and the pads according to all federal, state, and local regulations.

The coil should be cleaned only according to the procedure described above.

Accessories and Replaceable Components

Coil Replacement parts (FRU's - Field Replaceable Units) should be ordered by a GE Service Representative or a third party service provider.

If you do not have a GE Service Representative or third party service provider, please contact GE.

Table 4-2 1.5T CTL Coil Accessories and Replaceable Components:

Part Name	Part Number
4 and 8 Channel Ear Wedge Pads	E8800PE
4 Channel Patient Pad	E8800PD
8 Channel Pad Set	E8801SL

Contact Information

GE Healthcare Americas	
USA	800-558-5102
Canada	800-668-0732
GE Healthcare Global	
China	86-21-62192228
Taiwan	886-2-2505-7900
Singapore	65-291-8528
Australia	61-2-9975-5501
Japan	81-120-48-2630
Korea	82-31-740-6119
India	91-80-845-2923
Europe	33-1-41-19-76-76
GE Healthcare Manufacturer Information	
GE Website	www.gehealthcare.com
Manufacturer Address	GE Healthcare Coils 1515 Danner Drive Aurora, Ohio 44202-9273 USA
Manufacturer Phone Number	1-330-562-1000